



Menu

Mycophiles Attack! 🍄🔥

If you're a mushroom lover (a *mycophile*) you may have noticed an interesting mistake in the lecture on grammars – specifically with a portion of the prompt Dr. Brooks fed into the model:

"...

What is a mycelium?

A) Sporangia covered with colorful scales or ridges

B) Undifferentiated mass of branching filaments

C) The stem of the mushroom that carries spores from the fruiting body to the soil where it will grow and reproduce.

D) The cap of a mushroom

ANSWER: D"

This question, while correctly in the Aiken format, has an incorrect answer – only a level 1 mushroom forager would think mycelium refers to the cap of a mushroom instead of its undifferentiated mass of branching filaments! So how did this mistake come to be?

Well, aiming to automate more of his work, Dr. Brooks turned to a large language model to help generate some questions in order to demonstrate the few shots to be provided in this lecture. In this case, the model actually generated one syntactically correct question which was incorrect, and this wasn't caught until the lecture was filmed. So why not fix it before launching the course?

It turns out, this is a great example of some of the challenges large language model creators are facing with synthetic data generation. As discussed with Llama 2 code fine tuning, it is possible to use language model outputs as inputs to the data, either in the fine tuning process or (as was the case with the mushroom example) in-context training through few shot prompting. Errors in the data going into the model can increase the number of errors coming out of the model, a good example of the old adage "garbage in, garbage out".

So, does this mean you should not use generative AI for synthetic data generation? Not quite – synthetic data is an active area of research, and in many cases the use of synthetic data does actually improve the results of computational models, both in more traditional supervised machine learning tasks and in generative AI tasks. However, it is important to be aware of the potential limitations of synthetic data, as incorrect data may lead to suboptimal patterns being picked up on and emphasized on in the models you use. The cleaner the data you start with, the better the models will perform overall.

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